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Problem Set 2 Solutions

Enrico D. Turri[1]

Dynamic Programming: Computing Numerical Solutions

Question 1: Consider the Bellman equation for the social planning problem for the "standard model:"

$$v(k) = \max_{k'} \left\{ \log\left((1-\delta)k + Ak^{\alpha} - k'\right) + \beta v(k') \right\}.$$

Compute the value function and the policy function (the optimal choice of k' given k) using value function iteration on a discrete grid. Use the parameter values $\beta = 0.9$, $A = 3$, $\alpha = 0.3$, and $\delta = 0.2$. You can use $\bar{k} = 10$ as the upper bound for capital. Use $v_0(k) = 0$ for all k as your initial guess. Your capital grid should have at least 10,000 grid points.

1. Provide plots of the value and policy functions, and determine the steady-state level of capital.

   Answer: The code to obtain the value and policy functions is in the class notebook on Moodle, which also contains additional discussion. I plot them in Figure 1. The estimated steady-state level of capital is 4.560.

   As a sanity check, we can derive the steady state analytically from $k_{ss}^{\alpha-1} = (1/\beta - 1 + \delta)/(\alpha A)$, which, given the parameters, gives $k_{ss} = 4.5608$. Hence, the error is within one capital grid step.

2. Compute and plot time series for capital over 20 periods with starting values of $k_0 = 0.1$, $k_0 = 1$, and $k_0 = 10$.

   Answer: I plot the time series in Figure 2. From the time series, it is clear that convergence is quick, even if initial values are very different.

3. Comment on the implications of your results for the convergence of income levels across countries.

   Answer: Since convergence is quick, in this model, initial capital differences cannot explain the persistent differences between countries that we observe in the data.

[1] e.turri@lse.ac.uk

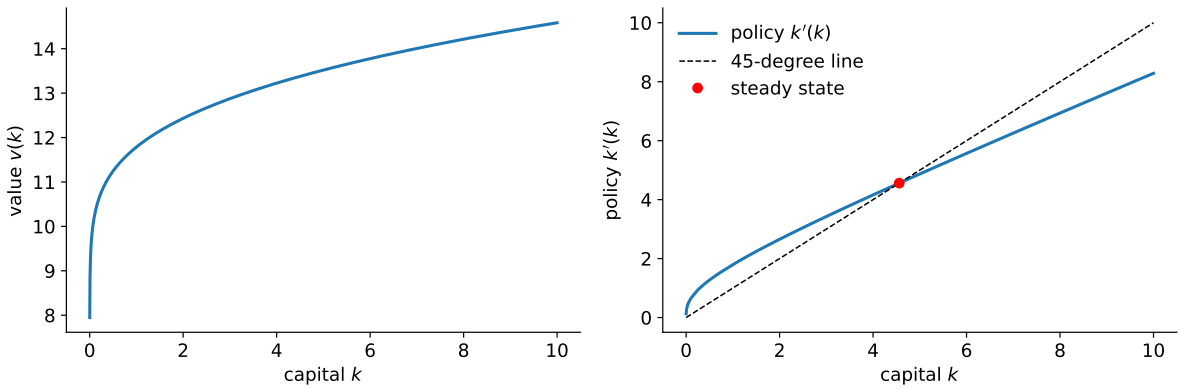


Figure 1: Value and policy function using value function iteration

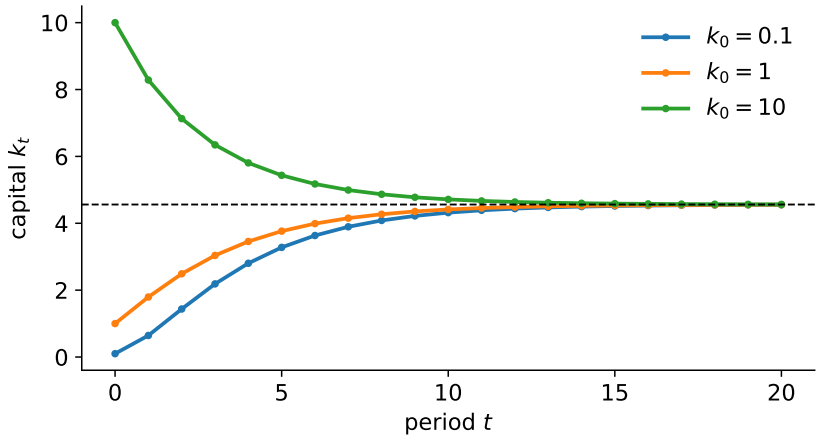


Figure 2: Time series of capital over 20 periods using different levels of k_0

Question 2: Redo your computation of the value function using the Howard policy improvement algorithm described in Ljungqvist and Sargent (2018). Briefly, this algorithm also iterates on the value function, but the policy function is only optimized infrequently (after the value function for a given policy has converged) rather than optimizing at every iteration. Use the same grids as above, use the initial guess $v_0(k) = 0$ for all k, and construct an optimal policy $k' = g_0(k)$ for all k corresponding to this value function (this is what you did in the first iteration above). Now, before optimizing the policy again, iterate on the value function with the *fixed* policy function $g_0(k)$ until convergence, using the criterion 10^{-5}. Call the resulting value function associated with this policy $v_1(k)$. Then, compute a new policy using $v_1(k)$ as the value function from tomorrow on. Compute the value $v_2(k)$ associated with this fixed policy function by iteration, taking $v_1(k)$ as the initial guess. Continue in this fashion until convergence of the value function. Verify that this algorithm gives the same outcome as standard value function iteration, and compare the required computation time for each algorithm. What explains the difference in computation time?

Answer: Figure 3 shows the value and policy function using this alternative solution method. The solution is the same as under the value function iteration method. Using Howard's policy improvement algorithm, the full simulation is around 10 times faster. The trade-off between the two methods is that value function iteration requires a (potentially costly) maximization operation for each iteration, while policy iteration maximizes only after each time the value function has converged given the policy. However, the total number of iterations tends to be larger in policy iteration (in this case, it is 450 for policy iteration, of which 11 maximizations and 439 policy evaluations, against 113 for value function iteration).

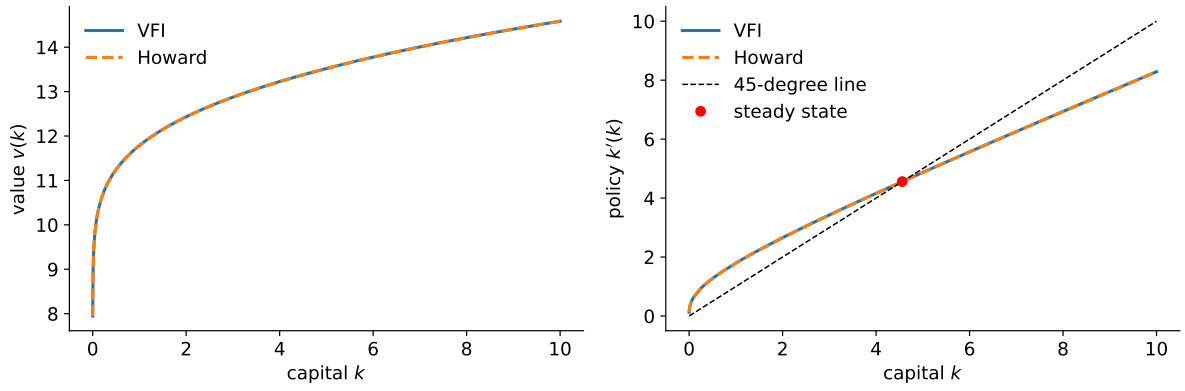


Figure 3: Value and policy function using Howard's policy improvement algorithm, compared with value function iteration

Dynamic Programming: Time to Build

Question 3: Consider the following economy. The representative consumer has preferences:

$$\sum_{t=0}^{\infty} \beta^t \log(c_t).$$

Output is produced using physical capital k_t. Current output can be transformed into consumption c_t or into investment i_t. It takes two periods for current investment to be transformed into productive capital. The resource constraints for the economy are therefore given by:

$$\begin{aligned} c_t + i_t &= Ak_t, \\ k_{t+2} &= i_t. \end{aligned}$$

The initial capital stocks for the first two periods k_0 and k_1 are given.

1. Formulate the social planning problem for this economy in sequence form.

   Answer: The planning problem is to choose sequences $\{c_t, k_{t+2}\}$ to solve

   $$\begin{aligned} \max_{\{c_t, k_{t+2}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \log(c_t) \\ \text{s.t.} \quad & c_t + i_t = Ak_t, \\ & k_{t+2} = i_t. \end{aligned}$$

2. Provide a Bellman equation that corresponds to the planning problem.

   Answer: The odd and even periods are separated; one can either formulate a Bellman equation that applies to odd/even periods only, or do both together. If both are done at the same time the Bellman equation is:

   $$v(k, k') = \max_{k''} \left\{ \log(Ak - k'') + \beta v(k', k'') \right\}$$

3. Conjecture that the value function is log-linear and that the policy function is linear, and solve for the value and policy functions.

   Answer: Using the log-linear guess we get:

   $$\begin{aligned} a_0 + a_1 \log(k) + a_2 \log(k') = \max_{k''} \Big\{ & \log(Ak - k'') \\ & + \beta(a_0 + a_1 \log(k') + a_2 \log(k'')) \Big\} \end{aligned} \tag{1}$$

The first-order condition for k'' and the envelope conditions for k and k' are:

$$\begin{aligned}\frac{1}{Ak - k''} &= \beta \frac{a_2}{k''},\\ \frac{a_1}{k} &= \frac{A}{Ak - k''},\\ \frac{a_2}{k'} &= \beta \frac{a_1}{k'}.\end{aligned}$$

The last condition implies $a_2 = \beta a_1$. The first two can then be written as:

$$\begin{aligned}\frac{1}{Ak - k''} &= \beta^2 \frac{a_1}{k''},\\ a_1 &= \frac{Ak}{Ak - k''}.\end{aligned}$$

Using the second condition to replace a_1 in the first gives:

$$\frac{1}{Ak - k''} = \beta^2 \frac{Ak}{k''(Ak - k'')},$$

which gives the policy function:

$$k'' = \beta^2 Ak.$$

Plugging this back into the second condition yields:

$$a_1 = \frac{1}{1 - \beta^2},$$

and consequently

$$a_2 = \frac{\beta}{1 - \beta^2}.$$

At last, plugging the policy function into (1) and setting $k = k' = 1$ to simplify gives:

$$a_0 = \log((1 - \beta^2)A) + \beta(a_0 + a_2 \log(\beta^2 A))$$

so that we have:

$$a_0 = \frac{1}{1 - \beta}\left(\log((1 - \beta^2)A) + \beta a_2 \log(\beta^2 A)\right).$$

Substituting $a_2 = \beta/(1 - \beta^2)$ and collecting the terms in $\log A$, whose coefficient is $1 + \beta^2/(1 - \beta^2) = 1/(1 - \beta^2)$, gives

$$a_0 = \frac{\log A + (1 - \beta^2)\log(1 - \beta^2) + \beta^2 \log(\beta^2)}{(1 - \beta)(1 - \beta^2)}.$$

The value function is therefore

$$v(k, k') = a_0 + \frac{\log(k) + \beta \log(k')}{1 - \beta^2},$$

with a_0 as above, and the policy function is $k'' = \beta^2 Ak$.

4. Compute the balanced growth rate of the economy as a function of parameters. Does the one-period consumption growth rate c_{t+1}/c_t converge to a constant from any initial condition?

   Answer: We have $k'' = \beta^2 Ak$, so the average growth rate per period is $1 + g = \beta\sqrt{A}$. The one-period growth rate does not converge in general. Since $c_t = Ak_t - k_{t+2} = (1 - \beta^2)Ak_t$, consumption grows at the same rate as capital, $c_{t+1}/c_t = k_{t+1}/k_t$. Even and odd periods are not linked: with $k_{t+2} = \beta^2 Ak_t$, the growth factor alternates forever between k_1/k_0 and $\beta^2 A\, k_0/k_1$. It is constant only if the initial conditions happen to satisfy $k_1/k_0 = \beta\sqrt{A}$.

Additional remarks

Splitting the problem into even and odd periods. Combining the two resource constraints, the planning problem is to choose $\{c_t, k_{t+2}\}_{t=0}^{\infty}$ to maximise $\sum_{t=0}^{\infty} \beta^t \log(c_t)$ subject to $c_t + k_{t+2} = Ak_t$ for all $t \geq 0$, given k_0 and k_1. Splitting the sum into even and odd periods,

$$\sum_{t=0}^{\infty} \beta^t \log(c_t) = \sum_{t=0}^{\infty} \beta^{2t} \log(c_{2t}) + \beta \sum_{t=0}^{\infty} \beta^{2t} \log(c_{2t+1}).$$

The first sum involves only even-dated variables and the second only odd-dated ones. The constraints split in the same way: in even periods, $c_{2t} + k_{2t+2} = Ak_{2t}$ involves only even-dated variables, and in odd periods, $c_{2t+1} + k_{2t+3} = Ak_{2t+1}$ involves only odd-dated ones. Since the objective is the sum of a function of the even-dated variables and a function of the odd-dated ones, and the constraints restrict the two groups separately, the maximum of the sum is the sum of the maxima:

$$\begin{aligned} v(k_0, k_1) &= \max_{\{c_t, k_{t+2}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^{2t} \log(c_{2t}) + \beta \sum_{t=0}^{\infty} \beta^{2t} \log(c_{2t+1}) \right\} \\ &= \max_{\{c_{2t}, k_{2t+2}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^{2t} \log(c_{2t}) + \beta \max_{\{c_{2t+1}, k_{2t+3}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^{2t} \log(c_{2t+1}) \\ &= v^{\text{even}}(k_0) + \beta v^{\text{odd}}(k_1), \end{aligned}$$

where each maximisation is subject to the constraints of its own group, and

$$v^{\text{even}}(k) = \max_{\{c_{2t},k_{2t}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \left(\beta^2\right)^t \log(c_{2t}) \quad \text{s.t. } c_{2t} + k_{2t+2} = Ak_{2t} \,,$$

$$v^{\text{odd}}(k) = \max_{\{c_{2t+1},k_{2t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \left(\beta^2\right)^t \log(c_{2t+1}) \quad \text{s.t. } c_{2t+1} + k_{2t+3} = Ak_{2t+1} \,.$$

The even and odd problems are identical, so $v^{\text{even}} = v^{\text{odd}}$. This is the AK model with logarithmic utility, with discount factor β^2 (one step of the chain is two periods) and gross return A. From the additional remarks to Question 3 of Problem Set 1, with β^2 in place of β and A in place of $1 - \delta + A$,

$$v^{\text{even}}(k) = v^{\text{odd}}(k) = b_0 + \frac{\log(k)}{1-\beta^2}, \qquad k'^{,\text{even}}(k) = k'^{,\text{odd}}(k) = \beta^2 Ak,$$

with

$$b_0 = \frac{\log A + (1-\beta^2)\log(1-\beta^2) + \beta^2 \log(\beta^2)}{(1-\beta^2)^2}.$$

Hence

$$v(k,k') = v^{\text{even}}(k) + \beta v^{\text{odd}}(k') = (1+\beta)b_0 + \frac{\log(k) + \beta\log(k')}{1-\beta^2},$$

which is the solution found in part 3: $a_1 = 1/(1-\beta^2)$, $a_2 = \beta a_1$, and $a_0 = (1+\beta)b_0$, since

$$\frac{1+\beta}{(1-\beta^2)^2} = \frac{1}{(1-\beta)(1-\beta^2)}.$$

The decomposition also explains why the policy function does not depend on k': k' belongs to the other group of periods, which does not interact with the choice of k''. Consistently, substituting $v(k,k') = v^{\text{even}}(k) + \beta v^{\text{odd}}(k')$ into the Bellman equation of part 2, and using $v^{\text{even}} = v^{\text{odd}}$, the term $\beta v^{\text{odd}}(k')$ cancels on both sides, leaving $v^{\text{even}}(k) = \max_{k''} \left\{\log(Ak - k'') + \beta^2 v^{\text{even}}(k'')\right\}$.

A Two-Sector Economy

Question 4: Consider an economy with a representative consumer whose preferences are described by:

$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}.$$

Assume that there are separate technologies for producing the consumption good and the investment good. Capital and labor can be moved freely across the two sectors.

The resource constraints are given by:

$$\begin{aligned} c_t &\leq l_{ct}^{\alpha} k_{ct}^{1-\alpha}, \\ i_t &\leq l_{it}^{\gamma} k_{it}^{1-\gamma}, \\ l_{it} + l_{ct} &\leq 1, \\ k_{it} + k_{ct} &\leq k_t, \\ k_{t+1} &= (1-\delta)k_t + i_t, \end{aligned}$$

with k_0 given.

1. Write down the sequential planning problem for this economy. What is the state variable?

   Answer: The sequential maximization problem, given k_0, is

   $$\begin{aligned} \max_{\{k_{t+1}, k_{ct}, l_{ct}\}_{t=0}^{\infty}} & \sum_{t=0}^{\infty} \beta^t \frac{\left(l_{ct}^{\alpha} k_{ct}^{1-\alpha}\right)^{1-\sigma}}{1-\sigma} \\ \text{s.t.} \quad & k_{t+1} = (1-\delta)k_t + (1-l_{ct})^{\gamma} (k_t - k_{ct})^{1-\gamma}, \\ & 0 \leq k_{ct} \leq k_t, \qquad 0 \leq l_{ct} \leq 1 \qquad \text{for all } t \geq 0. \end{aligned}$$

   From here, it is clear that the state variable is k_t.

2. Write down the Bellman equation for the recursive formulation of the planning problem.

   Answer: The Bellman equation is

   $$\begin{aligned} v(k) = \max_{k_c, l_c, k'} & \left\{ \frac{\left(l_c^{\alpha} k_c^{1-\alpha}\right)^{1-\sigma}}{1-\sigma} + \beta v(k') \right\} \\ \text{s.t.} \quad & k' = (1-\delta)k + (1-l_c)^{\gamma} (k - k_c)^{1-\gamma}, \\ & 0 \leq k_c \leq k, \qquad 0 \leq l_c \leq 1. \end{aligned}$$

3. Define a state space X that ensures that the utility function is bounded on the state space, and show that the operator defined by the Bellman equation satisfies Blackwell's sufficient conditions for a contraction.

   Answer: Capital cannot be negative, and it cannot grow without bound: even if all labor and capital produce investment goods, $k' \leq (1-\delta)k + k^{1-\gamma}$, which is smaller than k whenever $k > \delta^{-1/\gamma}$, the maximum sustainable level of capital. Let $\bar{k} = \max\{k_0, \delta^{-1/\gamma}\}$. Since $(1-\delta)k + k^{1-\gamma}$ is increasing in k and equals $\delta^{-1/\gamma}$ at $k = \delta^{-1/\gamma}$, we have $k' \leq \bar{k}$ whenever $k \leq \bar{k}$. Consumption then satisfies

$0 \leq c = l_c^\alpha k_c^{1-\alpha} \leq \bar{k}^{1-\alpha}$, so utility is bounded from above. The lower end of the state space depends on σ.

Case $\sigma < 1$. The utility function $u(c) = c^{1-\sigma}/(1-\sigma)$ is continuous on $[0, \infty)$, with $u(0) = 0$. Let $X = [0, \bar{k}]$, which is invariant, since $0 \leq k' \leq \bar{k}$ for any $k \in X$. On X,

$$0 \leq u(c) \leq \frac{\bar{k}^{(1-\alpha)(1-\sigma)}}{1-\sigma},$$

so utility is bounded on X.

Case $\sigma \geq 1$. Now $u(c) \to -\infty$ as $c \to 0$ (with $u(c) = \log(c)$ when $\sigma = 1$), so $[0, \bar{k}]$ does not work: at $k = 0$ consumption must be zero. Excluding zero is not enough either: on $(0, \bar{k}]$, $Tv(k)$ goes to $-\infty$ as $k \to 0$, so the sup norm of Tv is infinite. We therefore use a closed set bounded away from zero: $X = [\underline{k}, \bar{k}]$ with $0 < \underline{k} \leq k_0$, and require $k' \in X$. Utility is again bounded from above, so it is enough to find a strictly positive level of consumption that the planner can attain from every $k \in X$ while keeping $k' \geq \underline{k}$. Suppose the planner allocates half of capital and half of labor to each sector. Consumption is then $(1/2)^\alpha (k/2)^{1-\alpha} = k^{1-\alpha}/2 \geq \underline{k}^{1-\alpha}/2 > 0$, and next period's capital is $k' = (1-\delta)k + k^{1-\gamma}/2$, which is increasing in k, so that $k' \geq \underline{k}$ for all $k \in X$ if

$$\underline{k} \leq (1-\delta)\underline{k} + \frac{\underline{k}^{1-\gamma}}{2}, \qquad \text{that is,} \qquad \underline{k} \leq \frac{1}{(2\delta)^{1/\gamma}}.$$

The maximization in the Bellman equation can only do better than this particular allocation of capital and labor, so $Tv(k) \geq u(\underline{k}^{1-\alpha}/2) + \beta \inf v$ for all $k \in X$. Hence, with $0 < \underline{k} \leq \min\{k_0, (2\delta)^{-1/\gamma}\}$, the maximized value is bounded both from above and from below on X, even though utility is not. This works because, near zero, the investment technology is very productive relative to depreciation: $k^{1-\gamma}/k = k^{-\gamma} \to \infty$ as $k \to 0$.

In both cases, the Bellman equation maps bounded functions into bounded functions. Let $B(X)$ be the space of bounded functions $f : X \to \mathbb{R}$ with the sup-norm $d(f,g) = \sup_{x \in X} |f(x) - g(x)|$. The Bellman equation is then a mapping from $B(X)$ to $B(X)$. That is, the Bellman equation is an operator T such that

$$\begin{aligned} Tv(k) = \max_{k_c, l_c, k'} & \left\{ \frac{\left(l_c^\alpha k_c^{1-\alpha}\right)^{1-\sigma}}{1-\sigma} + \beta v(k') \right\} \\ \text{s.t.} \quad & k' = (1-\delta)k + (1-l_c)^\gamma (k - k_c)^{1-\gamma}, \\ & 0 \leq k_c \leq k, \qquad 0 \leq l_c \leq 1, \qquad k' \in X. \end{aligned}$$

Denote by $\Gamma(k)$ the set of choices (k_c, l_c, k') that satisfy these constraints, so that the maximization is over $(k_c, l_c, k') \in \Gamma(k)$. The solution is a fixed point of this mapping: $Tv(k) \equiv v(k)$. We now apply Blackwell's sufficient conditions.

Monotonicity: Let $f, g \in B(X)$ be such that $f(x) \geq g(x)\ \forall x \in X$ and define

$$(k_c^*(k), l_c^*(k), k^*(k)) \equiv \underset{(k_c,l_c,k')\in\Gamma(k)}{\arg\max} \left\{ \frac{\left(l_c^\alpha k_c^{1-\alpha}\right)^{1-\sigma}}{1-\sigma} + \beta g(k') \right\}.$$

Then,

$$\begin{aligned} Tf(k) &= \max_{(k_c,l_c,k')\in\Gamma(k)} \left\{ \frac{\left(l_c^\alpha k_c^{1-\alpha}\right)^{1-\sigma}}{1-\sigma} + \beta f(k') \right\} \\ &\geq \frac{\left((l_c^*(k))^\alpha (k_c^*(k))^{1-\alpha}\right)^{1-\sigma}}{1-\sigma} + \beta f(k^*(k)) \\ &\geq \frac{\left((l_c^*(k))^\alpha (k_c^*(k))^{1-\alpha}\right)^{1-\sigma}}{1-\sigma} + \beta g(k^*(k)) \\ &= Tg(k). \end{aligned}$$

Hence, monotonicity is satisfied.

Discounting:

$$\begin{aligned} T(f+a)(k) &= \max_{(k_c,l_c,k')\in\Gamma(k)} \left\{ \frac{\left(l_c^\alpha k_c^{1-\alpha}\right)^{1-\sigma}}{1-\sigma} + \beta \left(f(k') + a \right) \right\} \\ &= \beta a + \max_{(k_c,l_c,k')\in\Gamma(k)} \left\{ \frac{\left(l_c^\alpha k_c^{1-\alpha}\right)^{1-\sigma}}{1-\sigma} + \beta f(k') \right\} \\ &= Tf(k) + \beta a. \end{aligned}$$

Hence, discounting is also satisfied. This implies that T is a contraction mapping. Since $B(X)$ with the sup-norm is complete, the contraction mapping theorem implies that it has a unique fixed point $v^*(k)$.

Remark. With $\sigma \geq 1$, the lower bound $\underline{k}$ restricts the planning problem, so the solution on $X = [\underline{k}, \bar{k}]$ solves the original problem only if the constraint $k' \geq \underline{k}$ does not bind. This has to be checked afterwards: for instance, the optimal path of capital is monotone and converges to the steady state k_{ss}, so it never reaches a lower bound below $\min\{k_0, k_{ss}\}$. Alternatively, one can work on $(0, \bar{k}]$ and iterate on T starting from an upper bound of the value function, using the monotonicity of T instead of the contraction property (Stokey et al., 1989, Section 4.4).

4. Assume that the value function is differentiable, and derive and interpret the first-order and envelope conditions that characterize the optimal solution.

Answer: The first order conditions are

$$\begin{aligned}
\alpha l_c^{\alpha-1} k_c^{1-\alpha} \left(l_c^{\alpha} k_c^{1-\alpha}\right)^{-\sigma} - \lambda\gamma(1-l_c)^{\gamma-1}(k-k_c)^{1-\gamma} &= 0 \\
(1-\alpha) l_c^{\alpha} k_c^{-\alpha} \left(l_c^{\alpha} k_c^{1-\alpha}\right)^{-\sigma} - \lambda(1-\gamma)(1-l_c)^{\gamma}(k-k_c)^{-\gamma} &= 0 \\
\beta v'(k') - \lambda &= 0.
\end{aligned}$$

which simplify to

$$\begin{aligned}
\underbrace{\alpha l_c^{\alpha-1} k_c^{1-\alpha}}_{A} \underbrace{\left(l_c^{\alpha} k_c^{1-\alpha}\right)^{-\sigma}}_{B} &= \underbrace{\beta v'(k')}_{C} \underbrace{\gamma(1-l_c)^{\gamma-1}(k-k_c)^{1-\gamma}}_{D} \\
\underbrace{(1-\alpha) l_c^{\alpha} k_c^{-\alpha}}_{E} \underbrace{\left(l_c^{\alpha} k_c^{1-\alpha}\right)^{-\sigma}}_{B} &= \underbrace{\beta v'(k')}_{C} \underbrace{(1-\gamma)(1-l_c)^{\gamma}(k-k_c)^{-\gamma}}_{F}
\end{aligned}$$

From here, we see that A and E are the marginal product of labor and capital in the production of consumption and B is the marginal utility of consumption. On the right hand side, C is the marginal value of additional capital tomorrow and D and F are the marginal product of labor and capital in the production of the investment goods. Hence, we see that the FOCs dictate that labor and capital are optimally distributed over the production of consumption and investment goods.

The envelope condition is

$$v'(k) = \lambda\left(1-\delta+(1-\gamma)(1-l_c)^{\gamma}(k-k_c)^{-\gamma}\right)$$

which implies

$$v'(k) = \beta v'(k')\left(1-\delta+(1-\gamma)(1-l_c)^{\gamma}(k-k_c)^{-\gamma}\right).$$

This indicates that the marginal value of a unit of capital today is equal to the discounted value of a marginal unit of capital tomorrow times the marginal increase of capital tomorrow in response to a marginal increase in capital today.

In sum, we have three equations in three endogenous variables (k', k_c, l_c):

$$\begin{aligned}
\alpha l_c^{\alpha-1} k_c^{1-\alpha} \left(l_c^{\alpha} k_c^{1-\alpha}\right)^{-\sigma} &= \beta v'(k')\gamma(1-l_c)^{\gamma-1}(k-k_c)^{1-\gamma} \\
(1-\alpha) l_c^{\alpha} k_c^{-\alpha} \left(l_c^{\alpha} k_c^{1-\alpha}\right)^{-\sigma} &= \beta v'(k')(1-\gamma)(1-l_c)^{\gamma}(k-k_c)^{-\gamma} \\
v'(k) &= \beta v'(k')\left(1-\delta+(1-\gamma)(1-l_c)^{\gamma}(k-k_c)^{-\gamma}\right)
\end{aligned}$$

This allows us to solve for the optimal choice given a value function $v(k)$.

5. Show how you can use the conditions to find the steady state of the economy.

   Answer: Note that in steady state, we must have $k' = k$ such that the resource constraint implies

$$\delta k = (1 - l_c)^{\gamma}(k - k_c)^{1-\gamma}.$$

   Also, the first order conditions imply

$$\frac{\alpha k_c}{(1-\alpha)l_c} = \frac{\gamma(k - k_c)}{(1-\gamma)(1-l_c)}$$
$$1 = \beta\left(1 - \delta + (1-\gamma)(1-l_c)^{\gamma}(k-k_c)^{-\gamma}\right).$$

   From there, we have three equations in three unknowns.

Sequential-Markets Equilibria and Arrow-Debreu Equilibria

Question 5: Consider the standard optimal growth model represented by the planning problem:

$$\max_{\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t u(c_t, n_t) \right\}.$$

subject to:

$$c_t + k_{t+1} \leq (1-\delta)k_t + A(k_t)^{\alpha}(n_t)^{1-\alpha},$$
$$0 \leq n_t \leq 1,$$

with k_0 given.

1. Define a sequential-markets equilibrium.

   Answer: A sequential-markets equilibrium consists of prices $\{w_t, r_t\}_{t=0}^{\infty}$ and quantities $\left\{c_t^*, n_t^*, k_{t+1}^*\right\}_{t=0}^{\infty}$ such that

   - consumers maximize utility given the prices: $\left\{c_t^*, n_t^*, k_{t+1}^*\right\}_{t=0}^{\infty}$ solves

$$\max_{\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t, n_t)$$
$$\text{s.t.} \quad c_t + k_{t+1} \leq (1 - \delta + r_t)k_t + w_t n_t,$$
$$0 \leq n_t \leq 1,$$

     given k_0, and a no-Ponzi condition.
   - firms maximize profits: (k_t^*, n_t^*) solves

$$\max_{k,n} \left\{ Ak^{\alpha}n^{1-\alpha} - r_t k - w_t n \right\}$$

- the allocation is feasible

$$c_t^* + k_{t+1}^* \leq (1-\delta)k_t^* + A(k_t^*)^\alpha (n_t^*)^{1-\alpha}.$$

2. Define an Arrow-Debreu equilibrium.

   Answer: An Arrow-Debreu equilibrium consists of prices $\{p_t, w_t, r_t\}_{t=0}^\infty$ and quantities $\left\{c_t^*, n_t^*, k_{t+1}^*\right\}_{t=0}^\infty$ such that

   - consumers maximize utility given the prices: $\left\{c_t^*, n_t^*, k_{t+1}^*\right\}_{t=0}^\infty$ solves

$$\begin{aligned} \max_{\{c_t, n_t, k_{t+1}\}_{t=0}^\infty} & \sum_{t=0}^\infty \beta^t u(c_t, n_t) \\ \text{s.t.} \quad & \sum_{t=0}^\infty p_t \left(c_t + k_{t+1}\right) \leq \sum_{t=0}^\infty p_t \left((1-\delta+r_t)k_t + w_t n_t\right), \\ & 0 \leq n_t \leq 1, \end{aligned}$$

     given k_0, with the normalization $p_0 = 1$. No no-Ponzi condition is needed, since the household faces a single lifetime budget constraint.

   - firms maximize profits: (k_t^*, n_t^*) solves

$$\max_{k,n} \left\{ Ak^\alpha n^{1-\alpha} - r_t k - w_t n \right\}$$

   - the allocation is feasible

$$c_t^* + k_{t+1}^* \leq (1-\delta)k_t^* + A(k_t^*)^\alpha (n_t^*)^{1-\alpha}.$$

   The budget constraint is linear in k_{t+1}, which is multiplied by $p_t - p_{t+1}(1-\delta+r_{t+1})$. If this term were not zero, the household could attain infinite wealth by holding an arbitrarily large positive or negative amount of capital, which is not consistent with equilibrium. Hence, in equilibrium,

$$\frac{p_t}{p_{t+1}} = 1 - \delta + r_{t+1},$$

   a no-arbitrage condition. With it, all the k_{t+1} drop out of the budget constraint, which simplifies to

$$\sum_{t=0}^\infty p_t \left(c_t - w_t n_t\right) \leq (1-\delta+r_0)k_0.$$

3. Show that the two types of equilibria are equivalent. That is, for a given sequential-markets equilibrium, find the prices that implement the equilibrium allocation as an Arrow-Debreu equilibrium, and show that all equilibrium conditions are satisfied. Follow the same approach to show that any Arrow-Debreu equilibrium can be transformed into a sequential-markets equilibrium.

   Answer: The Euler equation in the sequential-markets equilibrium is such that

$$u_c(c_t, n_t) = \beta u_c\left(c_{t+1}, n_{t+1}\right)\left(1 - \delta + r_{t+1}\right)$$

   and that in the Arrow Debreu equilibrium is

$$u_c(c_t, n_t) = \beta u_c(c_{t+1}, n_{t+1})\frac{p_t}{p_{t+1}}.$$

   Clearly, if we introduce to the sequential market equilibrium a sequence of Arrow-Debreu prices $\{p_t\}_{t=0}^{\infty}$ such that $p_t/p_{t+1} = 1 - \delta + r_{t+1}$, the first order conditions coincide. The intratemporal condition $-u_n(c_t, n_t) = w_t\, u_c(c_t, n_t)$ is the same in both equilibria. Note also that by iterating on the budget constraint in the sequential equilibrium, we obtain

$$\sum_{t=0}^{\infty} \frac{c_t - w_t n_t}{\prod_{i=0}^{t}(1 - \delta + r_i)} + \lim_{t\to\infty} k_{t+1}\frac{1}{\prod_{i=0}^{t}(1 - \delta + r_i)} = k_0.$$

   Using the Euler equation, which implies

$$\prod_{i=1}^{t}(1 - \delta + r_i) = \frac{u_c(c_0, n_0)}{\beta^t u_c(c_t, n_t)},$$

   and the transversality condition $\lim_{t\to\infty} \beta^t u_c(c_t, n_t)\, k_{t+1} = 0$, we can derive that

$$\lim_{t\to\infty} k_{t+1}\frac{1}{\prod_{i=0}^{t}(1 - \delta + r_i)} = \lim_{t\to\infty} \frac{\beta^t u_c(c_t, n_t)\, k_{t+1}}{(1 - \delta + r_0)\, u_c(c_0, n_0)} = 0$$

   such that the budget constraint becomes

$$\sum_{t=0}^{\infty} \frac{c_t - w_t n_t}{\prod_{i=0}^{t}(1 - \delta + r_i)} = k_0.$$

   Using again that $p_t/p_{t+1} = 1 - \delta + r_{t+1}$ with $p_0 = 1$, so that $p_t = 1/\prod_{i=1}^{t}(1 - \delta + r_i)$, and multiplying by $1 - \delta + r_0$, the budget constraint becomes equivalent to that in the Arrow-Debreu equilibrium. Given that the budget constraint is the same and the Euler equation is the same, the sequential markets equilibrium is identical to the Arrow-Debreu equilibrium.

   Similarly, we can show that an Arrow-Debreu equilibrium can be transformed into a sequential-markets equilibrium by setting $p_t/p_{t+1} = 1 - \delta + r_{t+1}$. The Euler equation coincides. Also, feasibility combined with firm optimization imply the sequential budget constraint. Therefore, all the equilibrium conditions of the sequential markets equilibrium are satisfied.

References

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